

Memory-Efficient Test-Time Scaling and Latent Reasoning in Multilingual Language Models

The Tokenization Tax on Thinking: Latent vs. Explicit Test-Time Compute for Indic Reasoning Under Memory Constraints

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Problem: Test-time scaling (TTS) improves reasoning by sampling K solution paths, but every path carries its own KV cache, so memory grows with $K \times \text{trace length}$. On a 24 GB GPU this is the binding constraint. Indic scripts need roughly 2-4x more tokens than English for the same content, so Indic reasoning hits the memory wall 2-4x sooner. Multilingual models also reason through an English "pivot" in their middle layers, and this is weaker for low-resource languages. Latent reasoning (COCONUT) feeds hidden states back as "continuous thoughts" without decoding tokens, so its thinking cost is independent of tokenizer. No one has tested if it makes it a cheaper alternative to explicit search for languages that tokenize badly.

Research Question: RQ1 (pivot): When the vocabulary projection is removed from the reasoning loop, does the English pivot survive or dissolve? **RQ2 (tax):** How much does tokenization inflate the memory cost of explicit TTS in Hindi and Telugu/Kannada, and how much can KV quantization and probe-guided pruning recover? **RQ3 (bridge):** At matched peak memory, how does latent reasoning compare with explicit search across languages? Are continuous thoughts robust to quantization, and can a probe set the number of latent steps?

Approach:

(a) Novelty - Not Incremental: Strong at the intersection, moderate within individual pieces. To our knowledge, no prior work (i) trains an architecturally latent model multilingually, (ii) measures how tokenization inflates TTS memory across languages, or (iii) compares latent and explicit reasoning on an accuracy-vs-peak-memory axis. Existing pivot studies read internal states through the vocabulary projection, so none can answer what happens when that projection is absent from the loop. Taken alone, KV quantization (KIVI) and probe-based pruning (ReProbe) are incremental and heavily studied.

(b) Why It Is Publishable

- Every outcome is a finding: the pivot surviving or dissolving, a large or small tax, and a latent advantage or none all yield a result.
- It addresses gaps that recent papers state explicitly: necessity-first evaluation of latent reasoning, and memory as the main deployment barrier for TTS.
- It spans three communities (multilingual NLP, latent reasoning, efficient inference), which makes an identical project elsewhere unlikely.

Key References

Latent Reasoning

- Hao et al. (2024). *Training LLMs to Reason in a Continuous Latent Space (COCONUT)*. [arXiv:2412.06769](#)
- Geiping et al. (2025). *Scaling Test-Time Compute with Latent Reasoning (Huginn)*. [arXiv:2502.05171](#)

Multilingual Latent Language and the English Pivot

- Wendler et al. (2024). *Do Llamas Work in English? On the Latent Language of Multilingual Transformers*. [arXiv:2402.10588](#)
- Liu et al. (2026). *Large Reasoning Models Are (Not Yet) Multilingual Latent Reasoners*. [arXiv:2601.02996](#)

Test-Time Scaling and Memory

- Muennighoff et al. (2025). *s1: Simple Test-Time Scaling*. [arXiv:2501.19393](#)
- FastTTS (2026). *Accelerating Test-Time Scaling for Edge LLM Reasoning*. ASPLOS 2026. [arXiv:2509.00195](#)

KV-Cache Quantization and Lightweight Verification

- Liu et al. (2024). *KIVI: A Tuning-Free Asymmetric 2bit Quantization for KV Cache*. [arXiv:2402.02750](#)
- Ni et al. (2026). *ReProbe: Efficient Test-Time Scaling of Multi-Step Reasoning by Probing Internal States*. ACL 2026. [arXiv:2511.06209](#)

Data, Translation and Evaluation

- Gala et al. (2023). *IndicTrans2: Towards High-Quality and Accessible Machine Translation Models for all 22 Scheduled Indian Languages*. [arXiv:2305.16307](#)
- Shi et al. (2022). *Language Models are Multilingual Chain-of-Thought Reasoners (MGSM)*. [arXiv:2210.03057](#)

Application & Impact

Users: Teams deploying small reasoning models on consumer or edge GPUs, including Indian startups and public-sector or education tools serving Hindi- and Telugu-speaking users. Researchers in multilingual and efficient NLP are the second audience.

Where: Low-resource or on-device settings such as tutoring apps, regional-language assistants and offline deployments, where memory and cost are tight.

Impact: The project will deliver (1) a quantified tokenization-tax measurement that guides tokenizer and model choice for Indic languages, (2) practical guidance on whether to spend a fixed memory budget on explicit search or latent reasoning, (3) interpretability evidence on whether latent reasoning is English thinking in disguise, and (4) open-source code, translated reasoning data and an evaluation protocol.